Unit no. ° 15

CONSOLIDATION AND TRANSITION TO PREMIÈRE

25 May to 11 June 2027

I Review of calculation, functions and geometry

Example 15.1: Annual Review · Part A

1. Write 168 /252 in lowest terms; calculate 2 $^{-3} + 1/4$.

2. Factorise $x^2 - 6x$ + 9, then solve ($x - 3)^2 = 4$.

3. Solve $\dfrac{x-2}{x+1} \geq 0$.

4. For $f(x) = 4 - (x-1)^2$ on [−2; 3], give the maximum and the x-values where $f(x) = 0$.

5. With $A(0;1)$, $B(4;3)$, C(6; 4), prove that the points are collinear and give an equation of the line.

Solution:

1. $168/252 = 2/3$, and $2^{-3} + 1/4 = 1/8 + 2/8 = 3/8$. 2. $(x-3)^2$; $x - 3 = \pm 2$, so $S = \{1;5\}$. 3. Excluded -1, zero 2; signs + , −, +, so $S =]-\infty;-1[\cup[2;+\infty[$. 4. The square is positive or zero, so $f(x) \leq 4$, equality at x = 1. The zeros are $x = -1$, 3, both in the domain. 5. $\overrightarrow{AB} = (4;2)$, $\overrightarrow{AC} = (6;3) = \frac{3}{2}\overrightarrow{AB}$. An equation is $x - 2y + 2 = 0$.

Guidance for the teacher

Suggested mark scheme: 10 points, 2 per question. For question 3, reserve one point for the domain and the endpoints. For question 5, ask for the conclusion after the calculation.

II Review of data, probability and modelling

Example 15.2: Annual Review · Part B

1. Calculate the mean and median of 2; 4; 4; 6; 9. Determine Q_1, Q_3.
2. A price falls by 15%, then rises by 20%. Give the overall rate and the rate that would then be needed to restore the price.
3. In a fictional population of 100 people, 40 do sport (S), 30 come by bike (V), 18 do both. Build the two-way table and calculate $P_S(V)$, $P_V(S)$ for an equally likely draw.
4. For a rectangle with perimeter 24 m, express the area in terms of one dimension x and prove its maximum.

Solution:

1. Mean 25 /5 = 5, median 4. Quartile positions 2 and 4, so $Q_1 = 4$, $Q_3 = 6$.
2. Overall multiplier $0{,}85 \times 1{,}20 = 1{,}02$: a 2% increase. Reverse rate $1{,}02 - 1 = -1/51 \approx -1{,}96\,\%$.
3. Table: $S \cap V = 18$, $S \cap \overline{V} = 22$, $\overline{S} \cap V = 12$, $\overline{S} \cap \overline{V} = 48$. $P_S(V) = 18/40 = 45\,\%$, $P_V(S) = 18/30 = 60\,\%$.
4. $A(x) = x(12 - x) = 36 - (x - 6)^2$ on $]0; 12[$. The maximum is 36 m^2, attained for the square with side 6 m.

Guidance for the teacher

Suggested mark scheme: 10 points, respectively 2, 2, 3, 3. The review of parts A and B may be set as a one-hour test, then reviewed with the course materials.

III Choosing a strategy and preparing what's next

Example 15.3: A program to explain

For a given function f, explain the following program. What does it return for $f(x) = x^2$?

```
def tableau(f):
    valeurs = []
    for x in range(-3, 4):
        valeurs.append((x, f(x)))
    return valeurs
```

Solution:

It returns a list of pairs x-value/image for the integers of -3 to 3 inclusive : $(-3, 9), (-2, 4), (-1, 1), (0, 0), (1, 1), (2, 4), (3, 9)$. The bound 4 in `range` is excluded. The table does not describe the images of all real numbers between -3 and 3.

Six skills to use

Investigate : try out and organise approaches. **Model** : choose variables, domain and assumptions. **Represent** : move from the text to a table, a figure or a graph. **Reason** : justify and distinguish conjecture from proof. **Calculate** : choose a form and check the operations. **Communicate** : explain clearly with appropriate notations and units.

Example 15.4: End-of-year portfolio

Choose a successful piece of work and a corrected piece of work. For each, identify the skills used, explain a method and set a personal objective for the following year. Then redo an exercise corresponding to this objective without looking at the solutions.

Solution:

There is no single numerical answer. A satisfactory answer precisely identifies the problem, the chosen method, a justification and a check. The objective must be observable, by example: "state the forbidden values before solving a quotient equation", rather than " get better at calculation".

Guidance for the teacher

To present the Première pathways, draw on the courses actually offered in the school and on their current curricula. The booklet sets no automatic mark threshold for course choices.